

Helping Robots Succeed Despite Messy Human Data: Missingness in Multimodal Home-Health Sensing

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Abstract—In-home sensing is increasingly used to support aging in place and will likely play an important role in future robot-assisted health interventions. However, real-world home sensing data are often incomplete and noisy, and missing human data remains underexplored in robotics and human-robot interaction. In this work, we study missingness in a multimodal longitudinal in-home sensor dataset collected from $N = 100$ older adults over approximately 21 months, focusing on sleep-relevant data as a first probe. We examine how sleep data missingness was associated with other observed signals such as home occupancy and survey-derived context. This work can inform future missingness-aware validation layers for robot-supported home health sensing systems, where contextual signals can help determine whether missing sleep data should be ignored, flagged as uncertain, or investigated before downstream action.

I. INTRODUCTION

As the population of older adults increases, there has been growing interest in robots that can support aging in place. Social and service robots have the potential to help older adults remain independent by providing reminders, check-ins, and assistance with activities of daily living [1]. However, for robots to provide this support in everyday homes, they need reliable information about the person and their environment, which is a more complex sensing problem than most robot sensing and reasoning challenges to date. In-home sensing systems can prospectively provide this context by identifying normal behavior patterns and detecting deviations that may signal health decline, such as frailty, falls, or cognitive decline [2]–[4]. Yet more in situ studies are needed before robot-assisted care can be reliably deployed in homes, and these real-world deployments introduce challenges that are difficult to capture in controlled laboratory settings [5], [6]. One of these challenges is missing and unreliable sensor data, which becomes especially important when robots rely on sensed information to make decisions or perform actions [7]. The present work addresses this issue by examining the extent of missing human data, a challenge that is likely to grow as home robot deployments become more common (in scenarios like the one illustrated in Fig. 1).

Data loss is a common challenge in real-world in-home sensing. Wireless devices lose power, sensors fail, participants travel, people sleep outside the monitored environment, and health events such as hospitalization can interrupt data collection. In these settings, missingness is not only a technical problem; it can also reflect behavioral or contextual factors. More broadly, data are meaningfully “missing” when the unobserved values would have been relevant to the analysis or downstream use had they been observed [8].

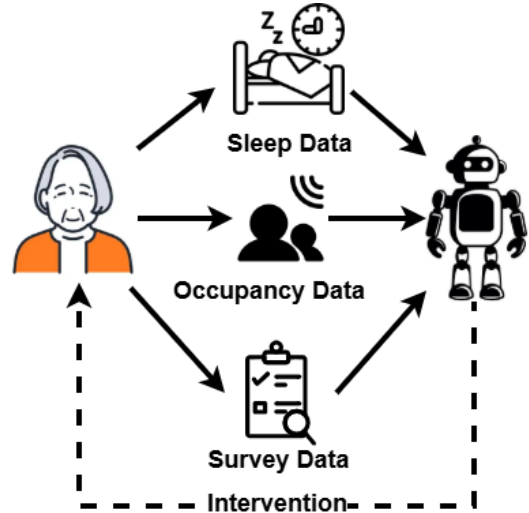


Fig. 1: Our envisioned multimodal in-home sensing pipeline, where sleep, occupancy, and survey data can inform prospective robot-assisted interventions for older adults.

For example, a night of missing bed-sensor data may reflect device failure, but it may also indicate that the participant was away from home, slept elsewhere, or experienced a change in routine. This issue is important for home-health monitoring and robot-assisted care for older adults, where in-home systems increasingly rely on long-term in-home sensing to support aging [9]. If such systems treat missingness inappropriately, they may produce misleading summaries or inappropriate prompts, alerts, or interventions. When sensed information is used to support caregiver decision-making or robot-mediated interventions for older adults, the validity of the data holds extra importance [10], [11]. More broadly, missingness in longitudinal in-home sensing may be related to observed context, such as occupancy patterns, survey reports, or time since enrollment, while also reflecting unobserved behavioral or health-related changes.

Prior work in smart-home and health-monitoring systems often acknowledges missingness, but typically addresses it using pre-processing strategies that are either field-specific or simple to implement, such as complete-case analysis, zero filling, or mean imputation [12]. These approaches may be reasonable for some downstream goals, but may also severely bias conclusions from downstream analyses when missingness is substantial [8]. Recent work has begun to address missingness and sensor-stream interpretation more directly, including methods for handling missing indoor sensor data and semantic annotation of sensor streams [13], [14].

However, weak standardization across homes, devices, and sensor placements remains a major challenge for comparing systems and generalizing findings [9], [13].

Bringing this idea back to robotics, in this paper, we examine missingness in longitudinal in-home sensing data from the Oregon Health & Science University (OHSU) Digital Evaluations and Technologies Enabling Clinical Translation for Alzheimer’s Disease (DETECT) study, which we use as an example dataset of the sort that could inform future in-home assistive robots. We focus on sleep-related missingness within a multimodal home monitoring environment. Our central research question is whether sleep data missingness showed systematic relationships with observed variables such as occupancy (home movement tracking) and survey context. Across the DETECT dataset, missing sleep data were substantial, motivating the need to better understand when missingness may reflect meaningful home context. We propose this work as a methodological contribution for in-home sensing systems and robotics because by proposing a missingness-aware validation layer for future assistive systems that will need to (among other challenges) decide whether sensed information is reliable enough to support intervention decisions in the home (Fig. 1). Rather than treating all missing sleep values uniformly, we propose to examine contextual factors associated with missingness and how multimodal context can support more informed missing-data handling. This is important for future robotic intervention systems that may need to decide whether sensed information is reliable enough to act on.

II. RELATED WORK

This paper sits at the intersection of two lines of work. The first concerns in-home sensing and robot-supported care for older adults, where multimodal home data may eventually guide prompts, alerts, or other interventions. The second concerns missing-data mechanisms, which provide a framework for reasoning about why sensor values may be absent and how those absences should be handled. Together, these areas motivate the present work, which treats missingness as both a sensing and a decision-support problem.

A. In-Home Sensing and Robot-Supported Care

In-home sensing systems for older-adult health monitoring measure broad behavioral signals such as activity, sleep, and mobility, as well as more contextual patterns such as movement through different areas of the home. By identifying routine behavior and detecting deviations that may indicate health decline, these systems can support independent living and aging in place. This motivation is especially important given the cost and scarcity of traditional care, the risk of falls and frailty, and the need to detect signs of cognitive or functional decline [2]–[4]. Overall, technical improvements in these systems aim to support independence for older adults while providing reassurance to caregivers and support networks.

Common sensing modalities include passive infrared (PIR) sensors for occupancy and movement, bed sensors for sleep

and restlessness, contact sensors for doors and cabinets, and survey or clinical measures that provide additional context. Early work from the Center for Advanced Studies in Adaptive Systems (CASAS) at Washington State University helped define smart-environment activity detection and activity-quality assessment [15]. Work from the Oregon Center for Aging and Technology similarly demonstrated the value of long-term, unobtrusive monitoring in older adults [16]. Related work from the University of Missouri showed how passive in-home sensing could support clinically meaningful health alerts rather than only activity recognition [17].

In-home sensing becomes especially relevant for robot-supported care when sensed context can inform a robot’s actions. Environmental sensing can help a robot estimate where a person is, infer routines, detect anomalies, or decide when an interaction should occur [1], [18], [19]. In this framing, the robot can serve either as an embodied interface for a broader home-monitoring system or as a localized intervention platform that uses a small number of external sensors to support a specific behavior or routine. Because much of this work is motivated by aging in place, recent research has explored health-promoting robots that use sensing beyond the robot itself to support older adults in home or home-like environments [5], [6].

But real-world deployment of sensing-supported robots remains limited. Long-term in-home evaluations are still relatively rare, with systems such as MoveCare and Robotic Activity Support (RAS) standing out as important in situ examples [20], [21]. Much of the existing literature remains feasibility-focused, leaving open questions about privacy, robustness, data quality, and long-term value. These questions are important when robot actions depend on personal health data. Poor or incomplete real-time data can lead to misleading summaries, false alarms, mistimed prompts, or loss of user trust.

This creates a need for validation methods that can assess the quality and context of in-home sensor data before the data is used for health interpretation or robot action. Validation prior to robot action could help determine whether missing data reflect sensor failure, expected absence, behavior change, or uncertainty. This would make downstream decisions easier to debug, justify, and communicate, which is important for health-support systems [22], [23]. In this paper, we focus on an upstream part of this broader problem by characterizing missing sleep data, examining whether missing values can be reconstructed, and evaluating how missing-data handling may shape analyses that could later inform caregiver-facing or robot-supported interventions.

B. Missingness Mechanisms

The reasons why the data are missing, or the missingness mechanism, informs how missing data should be handled. Introduced in [24], a missingness mechanism characterizes the dependence between the probability that a data point is missing and all observed and unobserved values of variables in the data. Missing data analysis prescribes a probabilistic

processes to both the values of the variables of interest *and* the binary indicators of missingness. Whether one can make substantive conclusions while neglecting to explicitly model the missingness mechanism (i.e. by fitting a logistic regression to the missingness indicators) depends on the missingness mechanism. The three missingness mechanisms are missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) [8]. Under MCAR, the probability that a value is missing is unrelated to both observed and unobserved data, so complete-case analysis can be valid. Under MAR, missingness depends only on values that are observed, which means inference may proceed without a missingness model (but not necessarily by doing dropping records with missing values). Under MNAR, missingness depends on unobserved values, making valid analysis more difficult because the missingness process must be modeled.

In home sensing, these distinctions are useful for determining whether missing data pose a problem for analysis. For example, if a random software failure causes a night of bed-sensor data to be lost, MCAR may be plausible, and records with missing values may be safely dropped from the analysis. If missing sleep data are more likely on nights when observed contextual signals indicate that the participant was away from home, MAR may be plausible, provided that, after conditioning on those observed signals, missingness no longer depends on the unobserved sleep value itself; under MAR missing values must be imputed. In contrast, if sleep values are more likely to be missing specifically when the participant is sleeping poorly in a way that is not otherwise captured by observed context, then the missingness is MNAR. MAR and MNAR missingness cannot be distinguished using observed data, but these categories provide a useful framework for interpreting missingness and motivating context-aware missing-data handling in home-health sensing.

In the context of robotic and health-support systems, decisions may need to be made from incomplete real-world sensing of human data. When sensed information is later used to guide prompts, alerts, or intervention decisions, the key question is whether the observed data are sufficient for an algorithm to act appropriately or whether missingness must first be modeled, contextualized, or clarified. This motivates our examination of sleep missingness, its relationship to multimodal home context, and the downstream consequences of different methods for handling missing data.

III. IN SITU DATASET

We used the OHSU DETECT dataset as an existing multimodal in-home sensing dataset to begin thinking about how data-informed robotic systems may operate in real homes. Below, we briefly describe the relevant characteristics of the dataset and then outline the subset of sensing streams and variables used in the present work.

TABLE I: Summary of expected, observed, and missing sleep-data nights in the demographic-complete analysis subset. Percentages for nights are calculated relative to expected participant-nights; percentages for participants are calculated relative to the number of participants.

Statistic	Count or value
Participants	100
Expected participant-nights	62,747
Observed sleep nights	35,201 (56.1%)
Missing sleep nights	27,546 (43.9%)
Participants with ≥ 1 missing night	100 (100.0%)
Participants with no missing nights	0 (0.0%)
Median observed nights per participant	336
Median missing nights per participant	283.5

A. Data Set Characteristics

The work draws on the DETECT-AD Dataset [25], [26], a longitudinal in-home sensing collection designed to capture behavioral and physiological changes in older adults living in their own homes. The cohort consists of $N = 100$ older adults (67% female; age $M=77.8$, $SD=6.62$) recruited to participate in a simulated prevention trial. The cohort is split between PET amyloid-negative ($n = 51$) participants and PET amyloid-positive ($n = 49$) participants, supporting comparison of “digital biomarker” trajectories across groups. Data streams available via the sensor system span mobility (gait speed, step counts, room transitions via an occupancy sensor array, a fitness tracker, and door sensors), cognition (computer device use, driving metrics via vehicle sensors), sleep (sleep times and efficiency via an under-mattress mat), socialization (dwell times, time out of home via area occupancy sensing), physiology (BMI via a biometric scale), and health behaviors (medication adherence via an electronic pillbox). The dataset is further distinguished by its relatively unobtrusive nature (no camera-based tracking), high dimensionality, and real-time continuous data collection, yielding ecologically valid in situ “digital biomarker” streams across these domains.

The passive sensor streams are complemented by weekly surveys in which participants self-report home absences, visitors, medication changes, falls, accidents, injuries, hospitalization, illness, change in life space or care needs, emotional well-being, pain, and interference in day-to-day activities [27], as well as monthly remote cognitive screening surveys (i.e., the Survey for Memory, Attention, and Reaction Time; SMART [28]). Annual in-person assessments provide additional ground truth via standardized cognitive and behavioral measures (e.g., MoCA [29], CDR [30]).

B. Dataset Construction for Analysis

The dataset used in this paper was constructed from a subset of the larger DETECT dataset so that one row of data represents one participant on a calendar date within the period of available sleep data. The presentation of data at the participant-day level was chosen so that nightly sleep missingness could be examined together with the home context available on the same date. The final analysis subset included

TABLE II: Descriptive comparison of nights with and without missing sleep data. N is the number of nights. Bed N, Home N, and Home D report mean event counts for bedroom at night, whole-home motion at night, and whole-home motion during the day, respectively. Away is the mean proportion of nights labeled away, and Days enr. is mean days since enrollment.

Miss?	N	Bed N	Home N	Home D	Away	Days enr.
Yes	29,239	94.2	348	1284	0.229	536
No	38,129	84.9	366	1304	0.088	420

$N = 100$ participants and 62,747 expected participant-night observations, corresponding to an average of approximately 627 nights, or about 21 months, per participant. Of these expected participant-nights, 35,201 nights had observed sleep data and 27,546 nights were missing, corresponding to 43.9% sleep-data missingness. Table I summarizes the observed and missing sleep-data nights in this subset.

For the present work, we focused on nightly sleep measures from the bed sensor as the primary target stream, daily occupancy features derived from in-home motion sensing as potential contextual and redundancy signals, survey-based context including away status and survey missingness to help interpret why sleep data may be absent, enrollment timing to capture longitudinal study effects, and selected (age and gender) participant-level demographic variables to account for between-person differences. The expected total count of nightly sleep observations were defined at the participant-day level for each calendar date between a participant’s first and last available bed-sensor sleep record. Within that window, `sleep_missing` was coded as 1 when the nightly bed-sensor sleep record was absent and 0 when it was present. Calendar features such as month and day of the week were derived from the date. This dataset served as the basis for the missingness and redundancy analysis presented in this manuscript.

IV. MISSINGNESS AND REDUNDANCY

We use multiple in-home sensing and survey streams to better understand sleep missingness. Our first analysis focuses on two related questions. *First*, we test whether missing sleep data are associated with other observed signals, which would rule out MCAR missingness. *Second*, we examine whether occupancy measures (night time bedroom/home events and day time home events), are useful predictors of observed sleep measures, thus enabling effective missing-data imputation.

A. Methods

We fit two types of mixed-effects models. The first is a logistic mixed model that predicts whether sleep data is missing on a given night using occupancy, survey context, and time-related variables, Equation 1. The second consists of linear mixed models that predict observed sleep outcomes from occupancy variables in order to evaluate the degree of overlap between the two data streams, Equation 2. These analyses are described in more detail in the Models section.

For both analyses, predictors were drawn from occupancy and survey data streams. Occupancy-derived predictors included nighttime bedroom events, nighttime home motion events, and daytime home motion events, each represented as a continuous count of PIR-detected activity. Nighttime was defined as 10:00 PM to 8:00 AM, with all remaining hours treated as daytime, consistent with related work [31]. Survey-derived predictors included *survey missing*, a binary indicator of whether the weekly survey was missing, and *away daily*, a binary indicator of whether the participant was reported as away from home during that week. Temporal covariates included calendar month, day of week, and days since enrollment to capture seasonal, weekly, and longitudinal trends.

The sleep outcomes analyzed in the redundancy models were duration in sleep, wake after sleep onset (WASO), awakenings, bed exit count, duration in deep sleep, and toss-and-turn count. These outcomes were selected to capture both overall sleep duration and more fragmented aspects of nighttime sleep. For the redundancy analysis, sleep outcomes and occupancy predictors were standardized before model fitting so that coefficients could be compared across variables measured on different scales. Binary context variables were coded as 0/1, and month and day of week were included as categorical factors. All models were fit in R using `lme4`, with `lmerTest` used for inference in the linear mixed-effects models.

B. Models

This analysis used two main model types. First, to study missingness, we fit a logistic mixed-effects model with a binary outcome indicating whether sleep data were missing or observed on a given night. Because nights were repeatedly observed for each participant, the model included a participant-specific random intercept. In contrast to the earlier descriptive comparisons, the missingness model included nighttime bedroom events, nighttime home motion events, and daytime home motion events together in the same model, while also adjusting for away-from-home status, survey missingness, month, day of week, and standardized time since enrollment. This allowed us to estimate the conditional relationship between each occupancy stream and sleep missingness while holding others constant. The goal of this model was to test whether sleep missingness was associated with observed home context.

The logistic mixed-effects model for missingness is:

$$\begin{aligned} \text{logit}(P(M_{it} = 1)) = & \beta_0 + \beta_1 B_{it} + \beta_2 H N_{it} + \beta_3 H D_{it} \\ & + \beta_4 A_{it} + \beta_5 S_{it} + \beta_6 M o_{it} \\ & + \beta_7 D O W_{it} + \beta_8 T_{it} + u_i, \end{aligned} \quad (1)$$

where M_{it} indicates whether sleep data were missing for participant i on night t , B_{it} is nighttime bedroom events, $H N_{it}$ is nighttime home motion events, $H D_{it}$ is daytime home motion events, A_{it} is away-from-home status, S_{it} is survey missingness, $M o_{it}$ is month of the year, $D O W_{it}$ is

TABLE III: Selected predictors of sleep-data missingness from a multivariable logistic mixed-effects model. Estimates are log-odds coefficients, and OR gives the odds ratio for a one-standard-deviation increase in the predictor. Intervals are 95% Wald confidence intervals.

Occupancy predictor	Est. (95% CI)	OR (95% CI)
Bedroom night events	0.023 (−0.009, 0.055)	1.02 (0.99, 1.06)
Home motion night events	−0.138 (−0.175, −0.102)	0.87 (0.84, 0.90)
Home motion day events	−0.357 (−0.390, −0.324)	0.70 (0.68, 0.72)
Away daily	0.937 (0.874, 1.001)	2.55 (2.40, 2.72)
Days since enrollment (z)	0.999 (0.975, 1.022)	2.71 (2.65, 2.78)

day of week, T_{it} is standardized time since enrollment, and u_i is a participant-specific random intercept.

Second, we fit linear mixed-effects models for each selected sleep outcome to study redundancy between occupancy and observed sleep measures. These models, which were fitted to complete cases, again included a participant-specific random intercept to account for repeated measurements within participant. Including bedroom events, nighttime home motion events, and daytime home motion events together in the same model allowed us to test whether any occupancy stream was predictive of the sleep outcomes while holding all else constant. The linear mixed-effects models for redundancy took the general form:

$$Y_{it} = \beta_0 + \beta_1 B_{it} + \beta_2 HN_{it} + \beta_3 HD_{it} + \beta_4 S_{it} + \beta_5 A_{it} + \beta_6 Mo_{it} + \beta_7 DOW_{it} + \beta_8 T_{it} + u_i, \quad (2)$$

where Y_{it} is the standardized sleep outcome for participant i on night t , B_{it} , HN_{it} , and HD_{it} denote the standardized occupancy predictors, S_{it} is survey missingness, A_{it} is away-from-home status, Mo_{it} is month of the year, DOW_{it} is day of week, T_{it} is standardized time since enrollment, and u_i is a participant-specific random intercept.

Before fitting the main models, we also performed descriptive comparisons between observed and missing nights to summarize differences in occupancy patterns, survey context, and time since enrollment.

C. Results

Descriptive comparisons between observed and missing sleep nights showed that missingness was not evenly distributed across the dataset. Missing sleep nights were more likely to coincide with days when a participant was away and with a greater total participation time in the study. Occupancy event counts also differed between observed and missing nights, with missing nights showing higher bedroom nighttime events but lower nighttime and daytime home motion counts. Together, these patterns suggest that missingness was related to an observable home context rather than a random noise variable (Table II).

The logistic mixed-effects missingness model supported the descriptive pattern that sleep missingness was related to home context, all else being constant. Higher daytime home motion and nighttime home motion were associated with lower odds of missing sleep data, *ceteris paribus*, with

TABLE IV: Selected occupancy associations with observed sleep outcomes from multivariable linear mixed-effects models. Estimates are standardized coefficients with 95% Wald confidence intervals.

Sleep outcome	Occ. predictor	Est. (95% CI)
Sleep duration	Home night events	−0.105 (−0.120, −0.090)
Deep sleep duration	Home night events	−0.088 (−0.105, −0.072)
Toss/turn count	Bed. night events	0.062 (0.051, 0.074)
Bed exit count	Bed. night events	0.053 (0.040, 0.067)
Awakenings	Bed. night events	0.045 (0.030, 0.059)
WASO duration	Bed. night events	0.044 (0.030, 0.059)

daytime home motion showing the strongest association. In contrast, bedroom nighttime events did not show a clear association with missingness after adjusting for the other occupancy measures. The full model also showed that away-from-home status and days since enrollment were strong predictors of missingness, with away nights showing substantially higher odds of missing sleep data and missingness increasing later in the study. Calendar effects were also present: September showed elevated odds of missingness relative to January, and weekend days showed higher odds of missingness than Monday. These temporal effects may reflect study-level data-collection patterns, seasonal gaps, or changes in participant routines such as travel or sleeping outside the monitored bed. For downstream intervention systems, this findings suggest that missingness itself may carry useful context about how sensed data should be interpreted before action (Table III). Full fixed-effects results for the missingness model are reported in Appendix I.

The redundancy analysis showed that the conditional relationships between occupancy event counts and observed sleep outcomes were generally modest. Across the tested outcomes, bedroom nighttime events showed the most consistent positive associations with sleep-disruption measures, including WASO, awakenings, bed exits, and toss-and-turn count, all else being equal. Nighttime home motion events was also informative, *ceteris paribus*, but its pattern differed by outcome: it was negatively associated with both sleep duration and deep sleep duration, all else held constant, suggesting that greater nighttime home activity may reflect sleep-relevant disruption or time spent out of bed. Even so, the standardized coefficients were generally small relative to the standardized outcome scale, with most estimates below 0.1 standard deviations in magnitude, indicating that occupancy variables should not be interpreted as strong direct substitutes for sleep outcomes. Instead, these results suggest that occupancy streams are more useful as contextual signals for understanding missingness and routine disruption than as one-to-one proxies for sleep measures (Table IV). Full redundancy-model results are reported in Appendix II.

V. DISCUSSION

The results of the present analysis suggest that occupancy and related context variables may play different roles in missing-data interpretation and future robot-supported sensing. Away status and time since enrollment were among

the strongest indicators of whether sleep data were missing, indicating that missingness reflected both participant context and broader longitudinal data-collection processes. Among the occupancy signals, daytime and nighttime home motion showed the clearest independent associations with whether sleep data were available, while bedroom nighttime events showed the clearest overlap with observed sleep-disruption measures. Even so, the redundancy relationships were generally small, meaning that occupancy should not be treated as a one-to-one proxy for sleep data. Instead, these variables are better interpreted as contextual signals for understanding missingness and supporting missingness-aware validation strategies. For roboticists, this suggests that ambient home context may be most useful not as a direct replacement for missing health data, but as additional evidence for deciding whether sensed information is reliable enough to support a prompt, check-in, or other intervention.

This work has several limitations. The paper does not evaluate a live robotic system or a concrete robot decision policy, so the robotics contribution is methodological rather than system-level. In addition, although missingness was related to survey and observed home context and was not consistent with MCAR, the current analysis cannot conclusively determine why any individual night was missing. Finally, the sensing setup does not fully capture sleep outside the monitored bed, which remains an important challenge for intervention-oriented home sensing systems.

A concrete next step would be to test these findings in a robot decision scenario, since away-related context was highly informative in our analysis but is only available retrospectively through weekly surveys. In that setting, a home robot could act not only as a consumer of multimodal data, but also as an additional sensing and uncertainty-resolution layer by helping gather missing context in real time and using it to decide whether to defer action, request clarification, or check on the person when an overnight sleep reading is missing.

VI. CONCLUSION

This paper examined missingness in multimodal longitudinal in-home sensing data from the OHSU DETECT study, with a focus on sleep-related missingness. We showed that missing sleep data were not consistent with completely random loss and were associated with observed home context such as occupancy, motion, and survey-derived indicators. These findings support treating missingness as an interpretable part of the sensing problem rather than only as a preprocessing nuisance.

For robotics and health-support systems, this missingness-aware validation layer is important because intervention decisions must often be made from incomplete and noisy real-world sensing streams, part of which originates from messy human data. Multimodal contextual information can help systems reason more carefully about missing data, retain more usable evidence, and support more stable models. Contextual signals can also help determine whether missing data are more likely to reflect expected absence, sensing

uncertainty, or a potential meaningful change in routine and therefore whether the available sensing evidence is trustworthy enough to support interventions. Although this work does not evaluate a live robot or a concrete intervention policy, this work hints that missingness will need to be treated as part of the sensing and decision-making problem itself, especially in intervention-oriented home monitoring settings.

ACKNOWLEDGMENTS

This work was supported in part by the NSF AI-CARING Institute (award number 2112633) and by the National Institutes of Health under award number R01AG078124. The authors thank Oregon Health & Science University (OHSU), Dr. Jeffrey Kaye, and Dr. Zachary Beattie for their support and for enabling access to the DETECT study data used in this work.

REFERENCES

- [1] B. Pradhan, D. Bharti, S. Chakravarty, S. S. Ray, V. V. Voinova, A. P. Bonartsev, and K. Pal, "Internet of things and robotics in transforming current-day healthcare services," *Journal of Healthcare Engineering*, vol. 2021, p. 9999504, 2021. [Online]. Available: <https://doi.org/10.1155/2021/9999504>
- [2] Centers for Disease Control and Prevention. (2024) Older adult falls data. Centers for Disease Control and Prevention. [Online]. Available: <https://www.cdc.gov/falls/data-research/index.html>
- [3] G. Kojima, "Frailty as a predictor of future falls among community-dwelling older people: A systematic review and meta-analysis," *Journal of the American Medical Directors Association*, vol. 16, no. 12, pp. 1027–1033, 2015.
- [4] N. L. Campbell, F. Unverzagt, M. A. LaMantia, B. A. Khan, and M. A. Boustani, "Risk factors for the progression of mild cognitive impairment to dementia," *Clinics in Geriatric Medicine*, vol. 29, no. 4, pp. 873–893, 2013.
- [5] A. Cruces, A. Jerez, J. P. Bandera, and A. Bandera, "Socially assistive robots in smart environments to attend elderly people—a survey," *Applied Sciences*, vol. 14, no. 12, 2024. [Online]. Available: <https://www.mdpi.com/2076-3417/14/12/5287>
- [6] A. Abou Allaban, M. Wang, and T. Padir, "A systematic review of robotics research in support of in-home care for older adults," *Information*, vol. 11, no. 2, 2020. [Online]. Available: <https://www.mdpi.com/2078-2489/11/2/75>
- [7] F. Pourpanah and A. Etemad, "Exploring the landscape of ubiquitous in-home health monitoring: A comprehensive survey," *ACM Trans. Comput. Healthcare*, vol. 5, no. 4, Oct. 2024. [Online]. Available: <https://doi.org/10.1145/3670854>
- [8] R. J. A. Little and D. B. Rubin, *Statistical analysis with missing data*, 2nd ed., ser. Wiley series in probability and statistics. Hoboken, NJ: Wiley, 2002.
- [9] P. P. Morita, K. Sahu, and A. Oetomo, "Health monitoring using smart home technologies: Scoping review," *JMIR mHealth and uHealth*, vol. 11, p. e37347, 2023.
- [10] G. Wilson, C. Pereyda, N. Raghunath, G. de la Cruz, S. Goel, S. Nesaie, B. Minor, M. Schmitter-Edgecombe, M. E. Taylor, and D. J. Cook, "Robot-enabled support of daily activities in smart home environments," *Cognitive Systems Research*, vol. 54, pp. 258–272, 2019. [Online]. Available: <https://doi.org/10.1016/j.cogsys.2018.10.032>
- [11] M. Luperto, J. Monroy, F.-A. Moreno, F. Lunardini, J. Renoux, A. Krpic, C. Galindo, S. Ferrante, N. Basilico, J. Gonzalez-Jimenez, and N. A. Borghese, "Seeking at-home long-term autonomy of assistive mobile robots through the integration with an iot-based monitoring system," *Robotics and Autonomous Systems*, vol. 161, p. 104346, 2023. [Online]. Available: <https://doi.org/10.1016/j.robot.2022.104346>
- [12] F. Palermo, Y. Chen, A. Capstick, N. Fletcher-Loyd, C. Walsh, S. Kouchaki, L. True, O. Balazikova, E. Soreq, G. Scott, H. Rostill, R. Nilforooshan, and P. Barnaghi, "Tihm: An open dataset for remote healthcare monitoring in dementia," *Scientific Data*, vol. 10, no. 1, p. 606, 2023.

- [13] K. Wuestney, D. Cook, C. Van Son, and R. Fritz, "Using indoor movement complexity in smart homes to detect frailty in older adults: Multiple-methods case series study," *JMIR Aging*, vol. 9, p. e77322, Jan. 2026.
- [14] W. Wu, J. M. Keller, M. Skubic, M. Popescu, K. R. Lane, and M. Rantz, "Semantic annotation of sensor data using a sequential possibilistic clustering methodology," in *2023 IEEE International Conference on Fuzzy Systems (FUZZ)*, 2023, pp. 1–6.
- [15] D. J. Cook and M. Schmitter-Edgecombe, "Assessing the quality of activities in a smart environment," *Methods of Information in Medicine*, vol. 48, no. 5, pp. 480–485, 2009.
- [16] J. Kaye, S. A. Maxwell, N. Mattek, T. L. Hayes, H. Dodge, M. Pavel, H. B. Jimison, L. Boise, and T. A. Zitzelberger, "Intelligent systems for assessing aging changes: Home-based, unobtrusive, and continuous assessment of aging," *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, vol. 66B, no. Suppl 1, pp. i180–i190, 2011.
- [17] M. Skubic, R. D. Guevara, and M. Rantz, "Automated health alerts using in-home sensor data for embedded health assessment," *IEEE Journal of Translational Engineering in Health and Medicine*, vol. 3, p. 2700111, 2015.
- [18] N. Ayari, A. Chibani, Y. Amirat, and E. Matson, "A semantic approach for enhancing assistive services in ubiquitous robotics," *Robotics and Autonomous Systems*, vol. 75, pp. 17–27, 2016, assistance and Service Robotics in a Human Environment. [Online]. Available: <https://doi.org/10.1016/j.robot.2014.10.022>
- [19] Y. Zhang, G. Tian, and H. Chen, "Exploring the cognitive process for service task in smart home: A robot service mechanism," *Future Generation Computer Systems*, vol. 102, pp. 588–602, 2020. [Online]. Available: <https://doi.org/10.1016/j.future.2019.09.020>
- [20] M. Luperto, J. Monroy, J. Renoux, F. Lunardini, N. Basilico, M. Bulgheroni, A. Cangelosi, M. Cesari, M. Cid, A. Ianes, J. Gonzalez-Jimenez, A. Kounoudes, D. Mari, V. Prisacariu, A. Savanovic, S. Ferrante, and N. A. Borghese, "Integrating social assistive robots, IoT, virtual communities and smart objects to assist at-home independently living elders: the MoveCare project," *International Journal of Social Robotics*, vol. 15, no. 3, pp. 517–545, 2023.
- [21] C. Pereyda, N. Raghunath, B. Minor, G. Wilson, M. Schmitter-Edgecombe, and D. J. Cook, "Cyber-physical support of daily activities: A robot/smart home partnership," *ACM Trans. Cyber-Phys. Syst.*, vol. 4, no. 2, Dec. 2019. [Online]. Available: <https://doi.org/10.1145/3365225>
- [22] Q. Xu, W. Xie, B. Liao, C. Hu, L. Qin, Z. Yang, H. Xiong, Y. Lyu, Y. Zhou, and A. Luo, "Interpretability of clinical decision support systems based on artificial intelligence from technological and medical perspective: A systematic review," *Journal of Healthcare Engineering*, vol. 2023, p. 9919269, Feb. 2023.
- [23] Q. Abbas, W. Jeong, and S. W. Lee, "Explainable ai in clinical decision support systems: A meta-analysis of methods, applications, and usability challenges," *Healthcare*, vol. 13, no. 17, 2025. [Online]. Available: <https://www.mdpi.com/2227-9032/13/17/2154>
- [24] D. B. Rubin, "Inference and Missing Data," *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976.
- [25] J. A. Kaye, J. Marcoe, A. B. Mar, E. J. Hanna, A. Pierce, L. C. Silbert, Y. Shang, D. Schwartz, S. Gothard, N. Mattek, W.-T. M. Au-Yeung, J. S. Steele, and Z. T. Beattie, "DETECT-AD (Digital Evaluations and Technologies Enabling Clinical Translation for Alzheimer's Disease): a simulated anti-amyloid clinical trial using digital biomarkers as primary outcome measures," *Alzheimer's & Dementia*, vol. 19, p. e071888, 2023.
- [26] J. A. Kaye, N. Mattek, A. B. Mar, J. Marcoe, E. J. Hanna, A. Pierce, Y. Shang, K. M. Wright, J. A. Wyatt, D. Schwartz, S. Gothard, W.-T. M. Au-Yeung, J. S. Steele, Z. T. Beattie, K. Duff, and L. C. Silbert, "DETECT-AD-Digital Evaluations and Technologies Enabling Clinical Translation for Alzheimer's Disease: Results of amyloid screening methodologies in a simulated anti-amyloid clinical trial using digital biomarkers as primary outcome measures," *Alzheimer's & Dementia*, vol. 20, p. e091154, 2024.
- [27] A. Pratap, E. C. Neto, P. Snyder, C. Stepnowsky, N. Elhadad, D. Grant, M. H. Mohebbi, S. Mooney, C. Suver, J. Wilbanks, L. Mangravite, P. J. Heagerty, P. Areán, and L. Omberg, "Indicators of retention in remote digital health studies: a cross-study evaluation of 100,000 participants," *NPJ digital medicine*, vol. 3, no. 1, p. 21, 2020.
- [28] M. I. Leese, N. Mattek, J. P. Bernstein, K. E. Dorociak, S. Gothard, J. Kaye, and A. M. Hughes, "The survey for memory, attention, and reaction time (smart): Preliminary normative online panel data and user attitudes for a brief web-based cognitive performance measure," *The Clinical Neuropsychologist*, vol. 37, no. 6, pp. 1257–1275, 2023.
- [29] Z. S. Nasreddine, N. A. Phillips, V. Bédirian, S. Charbonneau, V. Whitehead, I. Collin, J. L. Cummings, and H. Chertkow, "The montreal cognitive assessment, moca: a brief screening tool for mild cognitive impairment," *Journal of the American Geriatrics Society*, vol. 53, no. 4, pp. 695–699, 2005.
- [30] L. Berg, "Clinical dementia rating (cdr) ¹," *Psychopharmacology bulletin*, vol. 24, no. 3–4, p. 637, 1989.
- [31] L. Ji, M. L. Wallace, L. Master, M. M. Schade, Y. Shen, C. A. Derby, and O. M. Buxton, "Six multidimensional sleep health facets in older adults identified with factor analysis of actigraphy: Results from the einstein aging study," *Sleep Health*, vol. 9, no. 5, pp. 758–766, Oct. 2023.

APPENDIX I

FULL MISSINGNESS MODEL RESULTS

Table 1.1 Full fixed-effects results from the multivariable logistic mixed-effects model of sleep-data missingness. Estimates are log-odds coefficients; OR gives the odds ratio. Intervals are 95% Wald confidence intervals. Reference levels were January for month and Monday for day of week.

Predictor	Est. (95% CI)	OR (95% CI)
Intercept	-0.108 (−0.388, 0.172)	0.90 (0.68, 1.19)
Bedroom night events	0.023 (−0.009, 0.055)	1.02 (0.99, 1.06)
Home motion night events	-0.138 (−0.175, −0.102)	0.87 (0.84, 0.90)
Home motion day events	-0.357 (−0.390, −0.324)	0.70 (0.68, 0.72)
Away daily	0.937 (0.874, 1.001)	2.55 (2.40, 2.72)
Survey missing	0.080 (−0.028, 0.188)	1.08 (0.97, 1.21)
Month: Feb	-0.293 (−0.392, −0.195)	0.75 (0.68, 0.82)
Month: Mar	-0.766 (−0.864, −0.669)	0.46 (0.42, 0.51)
Month: Apr	-1.299 (−1.400, −1.199)	0.27 (0.25, 0.30)
Month: May	-1.300 (−1.399, −1.200)	0.27 (0.25, 0.30)
Month: Jun	-1.079 (−1.177, −0.981)	0.34 (0.31, 0.37)
Month: Jul	-1.536 (−1.641, −1.432)	0.22 (0.19, 0.24)
Month: Aug	-0.059 (−0.155, 0.038)	0.94 (0.86, 1.04)
Month: Sep	1.110 (1.011, 1.208)	3.03 (2.75, 3.35)
Month: Oct	0.200 (0.105, 0.295)	1.22 (1.11, 1.34)
Month: Nov	0.275 (0.181, 0.370)	1.32 (1.20, 1.45)
Month: Dec	-0.079 (−0.172, 0.015)	0.92 (0.84, 1.02)
DOW: Tue	-0.057 (−0.131, 0.016)	0.94 (0.88, 1.02)
DOW: Wed	-0.076 (−0.149, −0.002)	0.93 (0.86, 1.00)
DOW: Thu	-0.081 (−0.155, −0.008)	0.92 (0.86, 0.99)
DOW: Fri	-0.017 (−0.091, 0.056)	0.98 (0.91, 1.06)
DOW: Sat	0.184 (0.111, 0.258)	1.20 (1.12, 1.29)
DOW: Sun	0.254 (0.180, 0.328)	1.29 (1.20, 1.39)
Days since enrollment (z)	0.999 (0.975, 1.022)	2.71 (2.65, 2.78)

APPENDIX II

FULL REDUNDANCY MODEL RESULTS

Table II.1 Full fixed-effect coefficients from the multivariable linear mixed-effects redundancy models. Each column corresponds to a separate observed sleep outcome. Estimates are standardized coefficients with 95% Wald confidence intervals. Reference levels were January for month and Monday for day of week.

Predictor	Sleep duration	Deep sleep	Toss/turns	Bed exits	Awakenings	WASO
Intercept	0.069 (−0.044, 0.182)	0.052 (−0.040, 0.145)	0.118 (−0.071, 0.308)	0.031 (−0.125, 0.187)	0.012 (−0.126, 0.150)	0.009 (−0.139, 0.156)
Bedroom night events	0.025 (0.010, 0.041)	0.013 (−0.004, 0.030)	0.062 (0.051, 0.074)	0.053 (0.040, 0.067)	0.045 (0.030, 0.059)	0.044 (0.030, 0.059)
Home night events	−0.105 (−0.120, −0.090)	−0.088 (−0.105, −0.072)	−0.046 (−0.057, −0.035)	−0.019 (−0.032, −0.006)	−0.038 (−0.052, −0.024)	−0.029 (−0.043, −0.015)
Home day events	0.034 (0.019, 0.049)	0.018 (0.002, 0.034)	0.014 (0.004, 0.025)	0.035 (0.022, 0.048)	0.009 (−0.004, 0.023)	0.014 (0.001, 0.028)
Survey missingness	0.028 (−0.021, 0.077)	0.049 (−0.004, 0.102)	0.023 (−0.013, 0.058)	0.103 (0.061, 0.145)	0.006 (−0.039, 0.052)	0.036 (−0.009, 0.080)
Away status	−0.079 (−0.110, −0.048)	−0.078 (−0.112, −0.044)	−0.014 (−0.037, 0.009)	0.066 (0.039, 0.092)	0.039 (0.011, 0.068)	0.046 (0.018, 0.075)
Days since enrollment (z)	−0.044 (−0.053, −0.034)	−0.047 (−0.057, −0.036)	−0.028 (−0.035, −0.021)	0.019 (0.011, 0.028)	0.008 (−0.002, 0.017)	0.012 (0.003, 0.021)
Month: Feb	0.013 (−0.030, 0.056)	−0.031 (−0.078, 0.016)	−0.002 (−0.033, 0.029)	−0.024 (−0.061, 0.013)	−0.005 (−0.045, 0.035)	0.004 (−0.035, 0.044)
Month: Mar	−0.001 (−0.042, 0.040)	−0.021 (−0.066, 0.024)	−0.058 (−0.088, −0.028)	−0.031 (−0.067, 0.004)	−0.044 (−0.082, −0.005)	−0.029 (−0.066, 0.009)
Month: Apr	−0.124 (−0.165, −0.084)	−0.095 (−0.140, −0.051)	−0.102 (−0.132, −0.072)	−0.029 (−0.064, 0.006)	−0.028 (−0.066, 0.010)	−0.008 (−0.045, 0.029)
Month: May	−0.127 (−0.168, −0.086)	−0.092 (−0.136, −0.048)	−0.070 (−0.099, −0.040)	−0.060 (−0.095, −0.025)	−0.053 (−0.091, −0.015)	−0.037 (−0.074, −0.000)
Month: Jun	−0.120 (−0.162, −0.078)	−0.109 (−0.154, −0.064)	−0.061 (−0.091, −0.030)	−0.039 (−0.075, −0.004)	−0.005 (−0.044, 0.034)	−0.005 (−0.043, 0.033)
Month: Jul	−0.130 (−0.170, −0.089)	−0.082 (−0.126, −0.038)	−0.044 (−0.074, −0.015)	−0.040 (−0.075, −0.005)	−0.023 (−0.061, 0.014)	−0.005 (−0.042, 0.031)
Month: Aug	−0.171 (−0.216, −0.127)	−0.119 (−0.167, −0.071)	−0.060 (−0.092, −0.028)	−0.029 (−0.067, 0.009)	−0.014 (−0.056, 0.027)	0.008 (−0.032, 0.049)
Month: Sep	−0.100 (−0.150, −0.050)	−0.043 (−0.097, 0.011)	−0.023 (−0.059, 0.013)	−0.014 (−0.056, 0.029)	0.030 (−0.016, 0.076)	0.027 (−0.018, 0.072)
Month: Oct	−0.072 (−0.116, −0.028)	−0.024 (−0.072, 0.024)	−0.006 (−0.037, 0.026)	0.012 (−0.026, 0.049)	0.013 (−0.028, 0.054)	0.001 (−0.039, 0.041)
Month: Nov	−0.083 (−0.127, −0.039)	−0.027 (−0.075, 0.021)	−0.043 (−0.075, −0.010)	0.007 (−0.031, 0.045)	−0.002 (−0.043, 0.039)	0.010 (−0.030, 0.051)
Month: Dec	−0.073 (−0.114, −0.031)	−0.065 (−0.111, −0.020)	−0.011 (−0.042, 0.019)	−0.033 (−0.069, 0.003)	−0.039 (−0.078, −0.000)	−0.027 (−0.065, 0.011)
DOW: Tue	−0.030 (−0.061, 0.001)	−0.034 (−0.067, 0.000)	−0.001 (−0.023, 0.022)	0.034 (0.007, 0.061)	0.023 (−0.006, 0.051)	0.025 (−0.003, 0.054)
DOW: Wed	−0.001 (−0.033, 0.030)	−0.013 (−0.046, 0.021)	0.010 (−0.012, 0.033)	0.023 (−0.004, 0.050)	0.021 (−0.008, 0.050)	0.023 (−0.005, 0.052)
DOW: Thu	0.026 (−0.005, 0.057)	0.030 (−0.003, 0.064)	0.019 (−0.003, 0.042)	0.041 (0.014, 0.067)	0.041 (0.012, 0.070)	0.037 (0.008, 0.065)
DOW: Fri	0.053 (0.022, 0.085)	0.035 (0.001, 0.069)	0.008 (−0.015, 0.030)	0.029 (0.002, 0.056)	0.024 (−0.005, 0.053)	0.029 (0.001, 0.058)
DOW: Sat	0.014 (−0.018, 0.046)	0.029 (−0.006, 0.063)	0.002 (−0.021, 0.025)	0.033 (0.006, 0.061)	0.016 (−0.013, 0.046)	0.020 (−0.009, 0.049)
DOW: Sun	0.026 (−0.006, 0.058)	0.019 (−0.016, 0.054)	−0.006 (−0.029, 0.017)	0.024 (−0.004, 0.051)	0.012 (−0.017, 0.042)	0.022 (−0.007, 0.051)